

BCME (MEUC102)



B. Tech. 2nd Sem

PART - B

M.S Patel Department of Civil Engineering
CSPIT - Charusat

B Outline of the Course:



Sr. No.	Title of the Unit	Minimum nos. of hrs
<u>Part: A</u>		
1.	Fundamentals of Engineering Graphics	08
2.	Projections & Section of Solid	06
3.	Orthographic Projection	06
4.	Isometric Projections	06
5.	Computer Aided Drafting & Modeling	04
<u>Part: B</u>		
6.	Introduction of Mechanical Engineering	05
7.	Steam and Steam Generator	04
8.	Internal Combustion Engines	03
9.	Refrigeration and Air Conditioning Systems	03
<u>Part: C</u>		
10.	Scope of Civil Engineering	02
11.	Introduction to Surveying	06
12.	Elements of building Construction	07

UNIT: 07

SCOPE OF CIVIL ENGINEERING



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TOPICS TO BE COVERED

- Branches of Civil Engineering
- Scope of Civil Engineering
- Role of Civil Engineer

What is Engineering ???



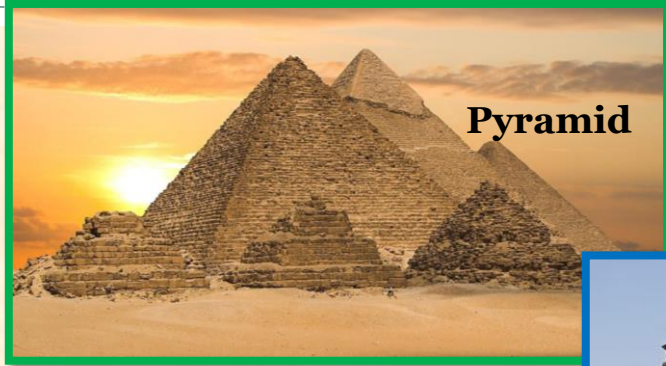
- Term Engineering is derive from the **Latin** word “**Ingenium**” meaning “**Cleverness**”
- Application of **scientific and mathematical knowledge** and rational thinking to improve living standards
- Engineer cleverly uses the **available natural resources** for the benefit of mankind
- Provides **comfort to mankind and makes life comfortable.**

Civil Engineering...

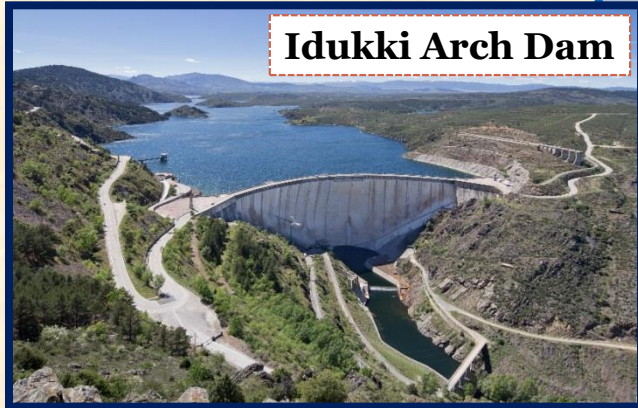


- **Civil Engineering** is the **oldest branch** of engineering.
- It is followed by mechanical engineering ,electrical engg., computer, electronics and so on
- It is core branch like mechanical and electrical engg.
- This professional engineering discipline deals with the **analysis, design, construction and maintenance of infrastructural facilities** such as buildings, bridges, dams, roads etc.

Marvels of Civil Engineering...



Pyramid



Idukki Arch Dam



Taj Mahal



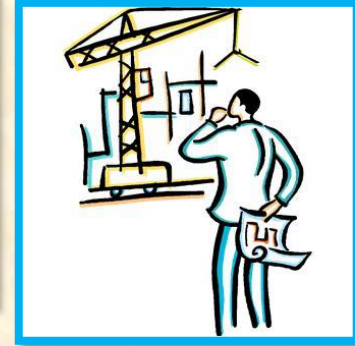
Burj Khalifa



Eiffel Tower

Branches of Civil Engineering

- ❖ Surveying & Levelling
- ❖ Construction Engineering
- ❖ Structural Engineering
- ❖ Geotechnical Engineering
- ❖ Water Resources Engineering
- ❖ Transportation Engineering
- ❖ Environmental Engineering



Surveying & Levelling

- Surveying includes measurements of **distances & angles** in **horizontal & vertical** planes
- Levelling is the measurement of **heights** in **vertical plane**
- **Chain, tape, compass & theodolite** instruments are used
- Used to **fix the alignment** of road, railway & canal
- **Electronic total station & Global Positioning System (GPS)** are modern electronic digital instruments for survey



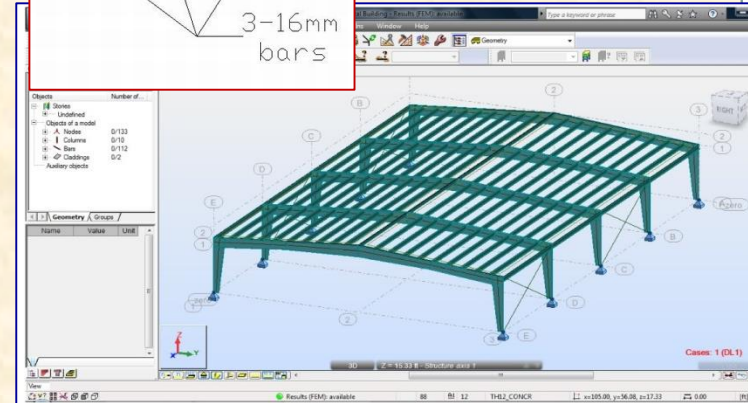
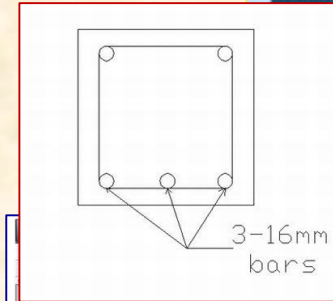
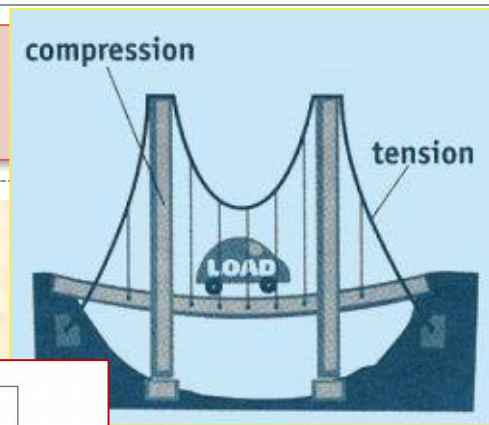
Construction Engineering

- Deals with designing, planning, construction and management of infrastructures such as roads, tunnels, bridges, buildings, dams etc.
- Comprises of men, material, time and money management.
- Importance will be on new construction practice, use of appropriate and local technology, safety of men and material, utilization of marginal materials etc.



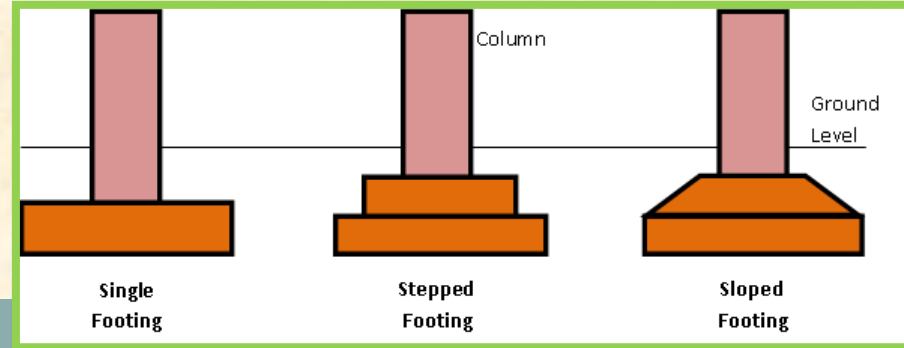
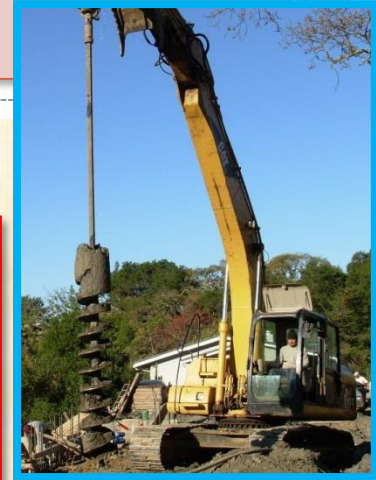
Structural Engineering

- Deals with structural analysis and design of structure
- Loads acting on various elements of a structure are calculated
- On the basis of loads and permissible stresses in materials, sections of structural elements i.e. beams, columns, slabs, footing etc. are decided
- Software like STADD PRO, STRUD, STRAP etc. are used for structural analysis and design



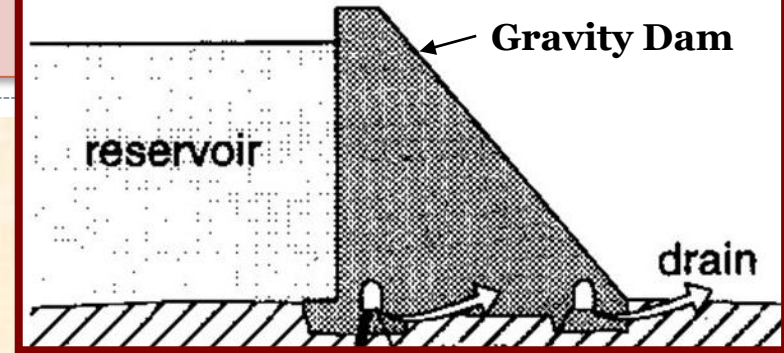
Geotechnical Engineering

- Deals with soil investigation and design of proper foundations of structure
- Soil properties changes from place to place
- It includes –
 1. Detailed soil investigation
 2. Bearing capacity determination
 3. Design of various type of foundations.



Water Resources Engineering

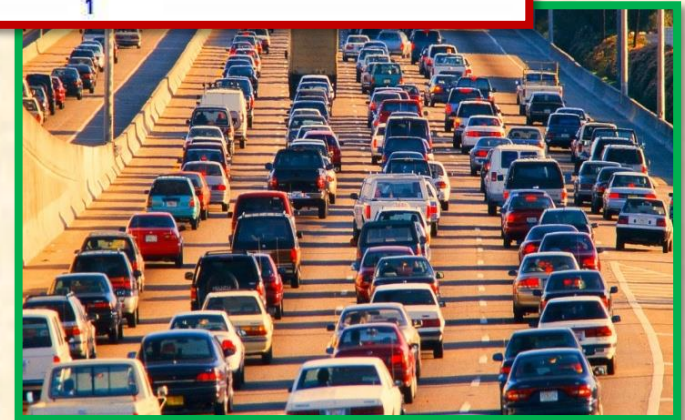
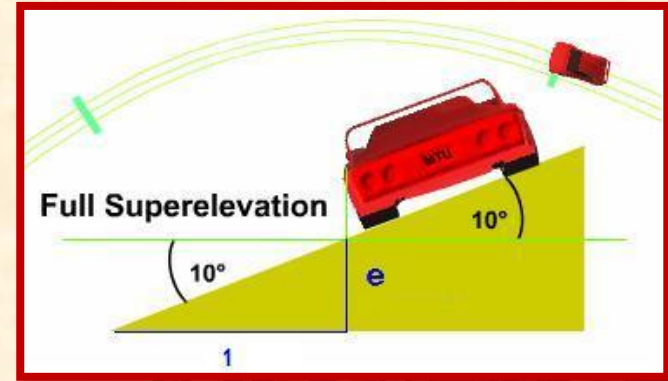
- Water resources engineering means utilization and development of water resources for agriculture, municipal & power generation purposes
- Includes Irrigation engineering, design of hydraulic structures like dam, canal etc.
- Deals with planning, designing and developing of water resources by constructing dam, hydropower station, canal, etc.
- Also includes water harvesting techniques, soil conservation and soil reclamation.



Spring
Irrigation

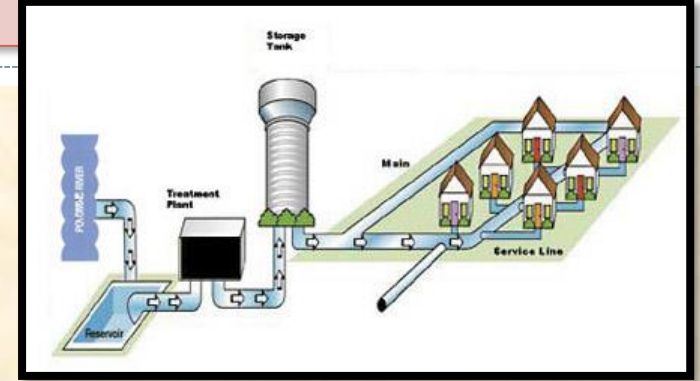
Transportation Engineering

- Involves movements of persons & goods from one place to another
- It has mainly three branches namely:
 1. Transportation way (highway, railway etc.) with its geometrical and structural design
 2. Traffic engineering which takes care of operation & control of vehicles
 3. Transportation planning



Environment Engineering

- Deals with pollution control and public health engineering
- Types of pollution are water, air, noise & other
- Public health engineering includes water treatment, waste water treatment, water distribution system and solid waste management
- Includes design, construction & maintenance of water treatment, water distribution system, sewage system



Scope of Civil Engineering



The main scope of civil engineering is :

- ❖ Planning
- ❖ Designing
- ❖ Estimating & Costing
- ❖ Supervising Construction
- ❖ Construction Management
- ❖ Maintenance of Structures like buildings, roads, bridges, dams etc.

Scope of Civil Engineering

Scope of Civil Engineering

According to Field of work, Area of services and Type of structure

- Building construction
- Construction of heavy structures
- Transportation engineering
- Water resources engineering
- Geo-technical engineering
- Environmental engineering
- Town planning

Function of Civil Engineering

- Surveying & levelling
- Planning
- Structural analysis & design
- Estimating, Costing, Valuation
- Construction Management
- Quality control and Monitoring

Scope According to Field of Work

1) Building Construction

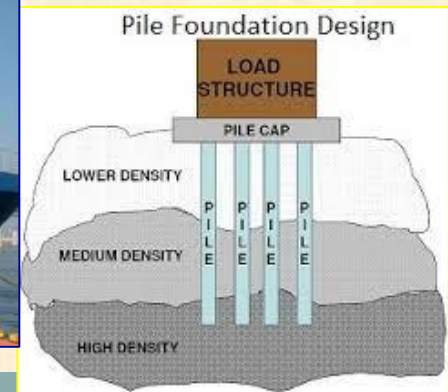
- **Constructing residential buildings** like tenements, flats, raw houses etc.
- **Public buildings** like schools, hospitals, shopping complexes etc.
- **Industrial buildings** like workshops, industrial sheds.
- **Includes study of building material & construction method.**



Scope According to Field of Work

2) Construction of Heavy Structures

- **Constructing dams, bridges, port, airport, tunnels, pile foundation etc. with advance construction techniques.**
- **Includes study of advance techniques, modern equipment and materials.**



Scope According to Field of Work

3) Geotechnical Engineering

- **Constructing** several type of foundations like simple footing, well foundation, pile foundation & foundations of machine subjected to vibrations
- **Includes** soil investigation and soil testing
- **Includes** construction of earthwork for highway & railway, earth dam.



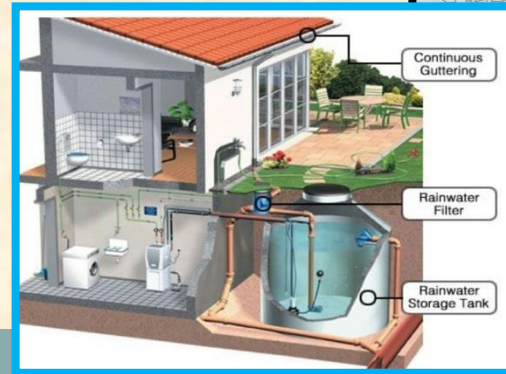
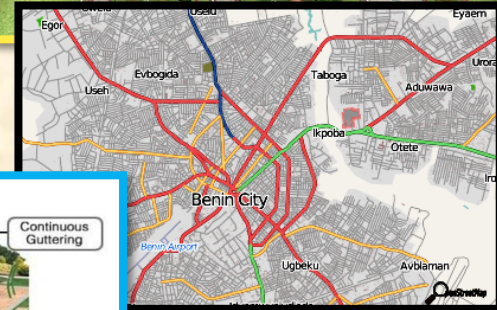
Scope According to Field of Work

4) Transportation Engineering

- **Constructing structures like roads, railway, bridges, tunnels, harbors, runway, airports.**
- **Also includes traffic engineering and study of highway materials.**



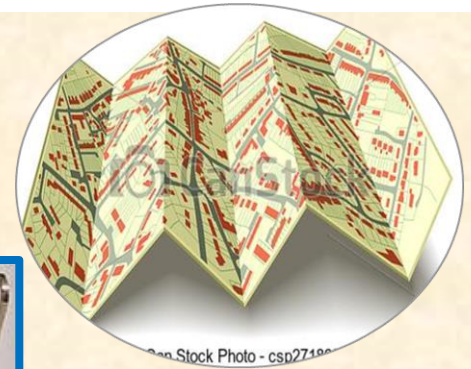
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Scope According to Functions

1) Surveying

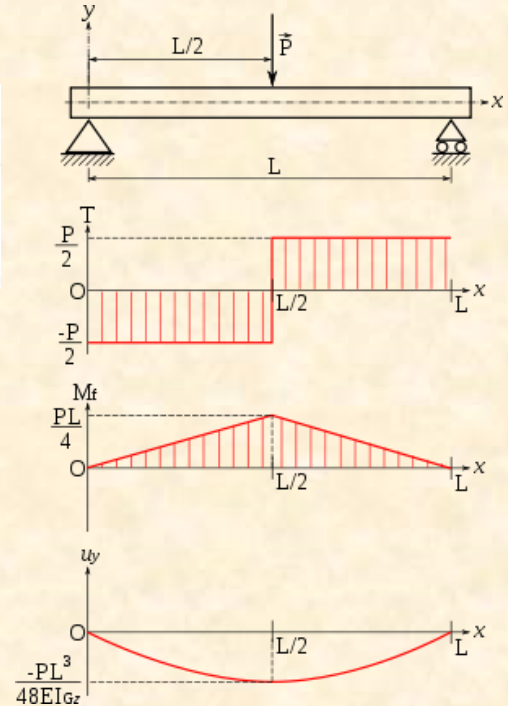
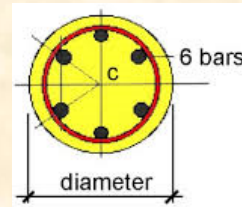
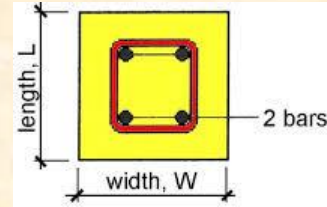
- To carry out surveying for setting out of works and for preparing map of land
- Leveling is carried out to measure the levels & to prepare contour map
- Measurement of distances & angles are taken



Scope According to Functions

2) Structural Analysis & Design

- To carry out structural design of the structure by selecting the type of material like concrete or steel
- By fixing the size & shape of various structural components like slab, beam, column etc.
- Structure analysis is required earlier to the designing of structure.



Scope According to Functions



3) Estimating, Costing, Valuation

- **Estimates** are prepared from **data of drawings, specifications, rates etc.**
- **Costing** is carry out to known the **actual expenditure in payment of bills to contractor & many other expenditure during construction work.**
- **Valuation** is carried out to know the **price or market value of property for the purpose of sale, purchases, loans, insurance.**

Construction Estimate Template

ESTIMATING & BUDGETING WORKSHEET (BuildingAdvisor.com)				
Description	Subcontractor or Vendor	Estimated Cost	Actual Cost	Variance
GENERAL REQUIREMENTS				
Plans and specifications				
Plan review				
Permits: Zoning, building, environmental, other				
Survey				
Impact fee				
Administrative costs				
Financing costs				
Legal fees				
Engineering fees				
Other fees				

Scope According to Functions

4) Quality Control

- To have a quality check by testing material and checking workmanship.
- During construction quality of material can be checked by testing various properties of materials
- Workmanship like dimensions, lines & levels, finishing etc. is also required to be checked



Role of Civil Engineer

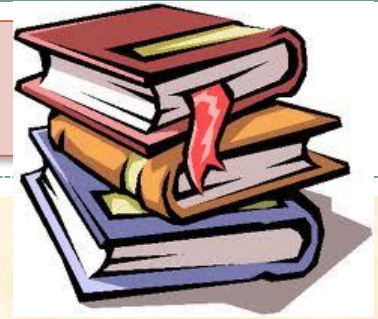
- To carry out **surveying and levelling**
- To fix the **alignments of roads, canals, railways, etc.**
- To carry out **detailed soil investigations**
- To carry out **planning of various types of building as per functional needs**
- To plan buildings according to **bye laws of local authorities**
- To carry out **Structural Analysis & design of various structures like building, bridges, dams, etc.**



Role of Civil Engineer

- He should also ensure that the **design is safe, durable and economical**
- To **supervise the work** during execution, to ensure the progress of work, to control the quality of work
- To carry out **testing of civil engineering materials** like cement, sand, aggregates, reinforcement bars, soil, etc.
- To carry out **valuation of land of building** for the purpose of sale/buy/taxation

References...



- Khasia R.B. and Shukla R.N., “Elements of Civil Engineering”, Mahajan Publication.
- Kandya Anurag, “Elements of Civil Engineering”, Charotar Publishing House Pvt. Ltd.